

Question

Salt **A** is a white powder with water of crystallization and plays an important role in the leather industry and dyeing and printing industries. 1.260 g of salt **A** was heated to 150°C in a N₂ atmosphere, losing 34.6% of its weight to obtain **B**, corresponding to the loss of all water of crystallization; **B** was further heated to constant weight, releasing a mixture of two gases in a molar ratio of 1 : 1, with a density of 0.61 g/L(298.15 K, 100kPa); the remaining solid was a white powder **C**, with a mass of 0.642 g. The solutions obtained by dissolving **A**, **B**, and **C** in water all have a certain alkalinity. Which of the following options is correct?

- A.** All other options are incorrect
- B.** A mixed gas does not necessarily contain H_2 .
- C.** Both gases released during thermogravimetric analysis are flammable.
- D.** If the reaction is carried out in an air atmosphere, the temperature at which **C** is formed will be higher.
- E.** Salt **C** can be used as a leavening agent alone in baking.
- F.** Salt **A** contains 1 crystal water.

Answer

Correct Answer: C

Detailed Explanation

The molar mass of the mixed gas is: $M = \rho RT/p = 15.0 \text{ g/mol}$,

CHECKPOINT

1 PTS

Therefore, the mixed gas must contain H_2 .

Since the molar ratio of the two gases is 1 : 1, the molar mass of the other gas is 28 g/mol,

CHECKPOINT

1 PTS

Which is CO.

$$n(\text{H}_2) = n(\text{CO}) = [1.260 \text{ g} \times (1 - 0.346) - 0.642 \text{ g}] / [2 \times 15.0 \text{ g/mol}] = 6.07 \times 10^{-3} \text{ mol}$$

The number of moles of water of crystallization is $n(\text{H}_2\text{O}) = 1.260 \text{ g} \times 0.346 / 18.0 \text{ g/mol} = 2.42 \times 10^{-2} \text{ mol}$

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1 PTS

The number of moles of water of crystallization is calculated

$$n(\text{H}_2\text{O}) = 4n(\text{H}_2) = 4n(\text{CO})$$

Suppose there are 2 waters of crystallization, then $M = 2m/n(\text{H}_2\text{O}) = 104 \text{ g/mol}$ minus 2 H_2O , $1/2 \text{ CO}$ and $1/2 \text{ H}_2$,

CHECKPOINT

1 PTS

It can be concluded that A is $\text{HCOONa} \cdot 2\text{H}_2\text{O}$

The molar mass of the mixed gas is: $M = \rho RT/p = 15.0 \text{ g/mol}$, The small molecule gases are H_2 , N_2 , O_2 , etc., but only H_2 has a molar mass within 15.0 g/mol , so it must contain H_2 .

CHECKPOINT

1 PTS

B is incorrect.

The two gases released by thermogravimetry are H_2 and CO , both of which are flammable.

CHECKPOINT

1 PTS

C is correct.

If the reaction is carried out in an air atmosphere, the generated gases are all reducing and can react with O_2 . This reaction is coupled with the decomposition reaction of sodium formate.

CHECKPOINT

1 PTS

The temperature for generating **C** will be lower. D is incorrect.

Salt **C** is Na_2CO_3 , which requires the addition of acid to produce H_2O and CO_2 . Sodium bicarbonate (commonly known as baking soda) has this unique effect that sodium carbonate (commonly known as soda ash) does not have, that is, it can independently generate gas as a leavening agent without relying on acidic substances to trigger the reaction. It can be used alone as a leavening agent in baking.

CHECKPOINT

1 PTS

Therefore, E is incorrect.

Calculated,

CHECKPOINT

1 PTS

Salt **A** contains 2 waters of crystallization. F is incorrect.